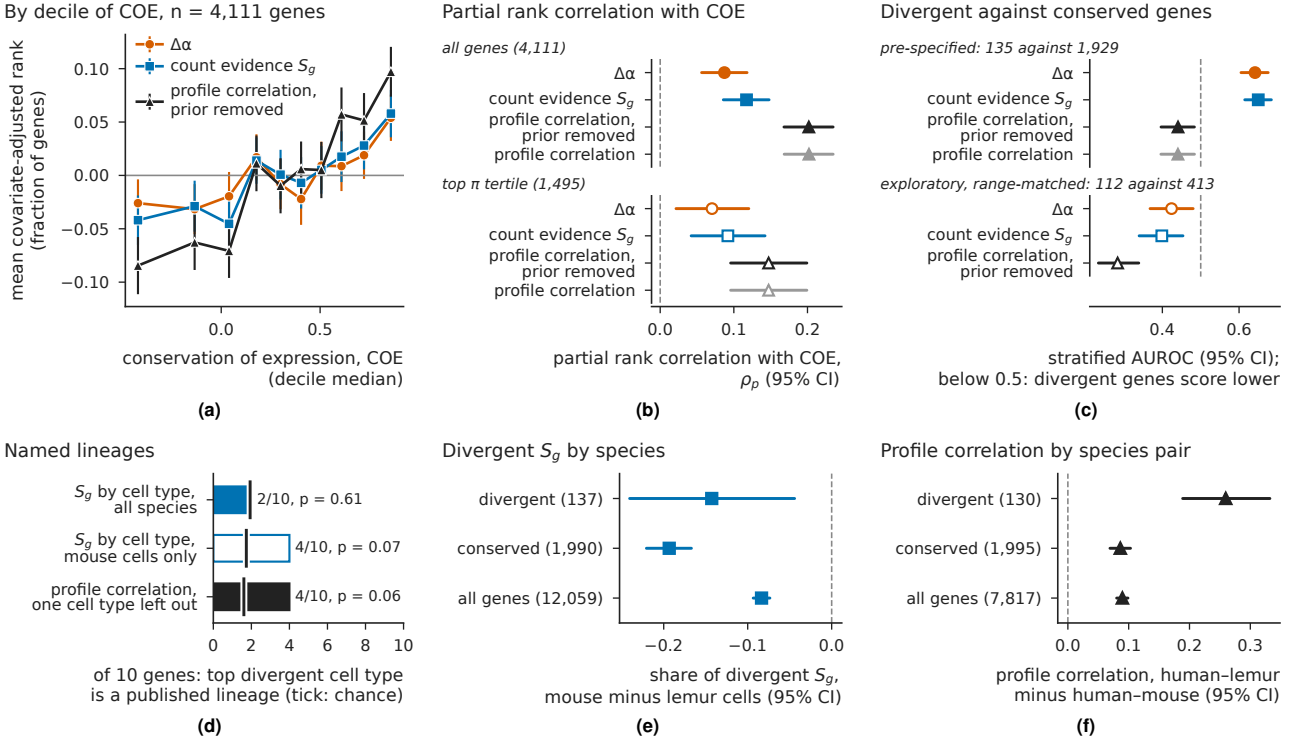

By decile of COE, n = 4,111 genes
Δα
count evidence S_g
profile correlation, prior removed
mean covariate-adjusted rank (fraction of genes)
0.10
0.05
0.00
−0.05
−0.10
0.0
0.5
conservation of expression, COE (decile median)
(a)
Partial rank correlation with COE
all genes (4,111)
Δα
count evidence S_g
profile correlation, prior removed
profile correlation
top π tertile (1,495)
Δα
count evidence S_g
profile correlation, prior removed
profile correlation
0.0
0.1
0.2
partial rank correlation with COE, ρ_p (95% CI)
(b)
Divergent against conserved genes
pre-specified: 135 against 1,929
Δα
count evidence S_g
profile correlation, prior removed
profile correlation
exploratory, range-matched: 112 against 413
Δα
count evidence S_g
profile correlation, prior removed
0.4
0.6
stratified AUROC (95% CI); below 0.5: divergent genes score lower
(c)
Named lineages
S_g by cell type, all species
2/10, p = 0.61
S_g by cell type, mouse cells only
4/10, p = 0.07
profile correlation, one cell type left out
4/10, p = 0.06
0
2
4
6
8
10
of 10 genes: top divergent cell type is a published lineage (tick: chance)
(d)
Divergent S_g by species
divergent (137)
conserved (1,990)
all genes (12,059)
−0.2
−0.1
0.0
share of divergent S_g, mouse minus lemur cells (95% CI)
(e)
Profile correlation by species pair
divergent (130)
conserved (1,995)
all genes (7,817)
0.0
0.1
0.2
0.3
profile correlation, human–lemur minus human–mouse (95% CI)
(f)